

SAI SATWIK BIKUMANDLA

Albany, New York, United States | bikumandlasaisatwik@gmail.com | (518) 614-1904 | [GitHub](#) | [LinkedIn](#) | [Website](#)

PROFESSIONAL SUMMARY

Full-stack Software Engineer with hands-on experience designing and shipping production web applications end-to-end — from requirements and system design through REST API development, microservices, cloud deployment, and testing. Comfortable owning the full SDLC, collaborating across cross-functional teams, and translating business requirements into working software. Experienced with Docker, Kubernetes, and both AWS and GCP cloud services; actively deepening enterprise Java (Spring Boot) and Angular expertise to complement a strong foundation in API design, microservices architecture, and full-stack delivery.

SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, C++
- **Full-Stack & API:** REST API Design, Microservices Architecture, Distributed Systems, Node.js, React.js, Next.js, Flask, FastAPI, JWT/NextAuth, System Design
- **Cloud & DevOps:** AWS, Google Cloud Platform, Docker, Kubernetes, CI/CD Pipelines, Linux, systemd, Database Optimization
- **Databases:** MongoDB, MySQL, Google Cloud SQL
- **AI & LLMs:** Local LLM inference (Ollama, Qwen3), structured JSON output, prompt design, TensorFlow/Keras, NLP
- **Currently Building:** *Spring / Spring Boot, Angular — actively expanding enterprise Java stack alongside existing REST API and microservices experience*
- **Practices & Tools:** Git, Code Review, Unit Testing, Agile/Scrum, Technical Documentation, Requirements Elicitation
- **Spoken Languages:** Telugu (Native), Hindi (Fluent), English (Fluent), German (Actively Learning)

RELEVANT PROJECTS

Job Agent | Self-Hosted AI Job Search Assistant (Python, FastAPI, Ollama, Linux)

- Built a job-search agent that runs on a CPU-only Linux server with local LLMs and no cloud APIs: each night it reads ~13,000 postings from ~140 company career boards (Greenhouse, Lever, Ashby, Workday, SmartRecruiters APIs), extracts required skills into a fixed JSON schema, and scores fit in code.
- Designed a human-in-the-loop pipeline that prepares tailored resumes, cover letters, and form answers overnight; a keyboard-first FastAPI web panel lets me approve each application, and a browser userscript fills and submits only the approved ones.
- Secured it with least privilege (separate Unix users joined by one sudo rule, firewall rules against self-approval, tailnet-only access, read-only IMAP), and made it self-deploy via systemd with automatic rollback; prompt caching cut per-step prompt processing from ~30 s to 1.7 s.

LaTeX Resume Builder | Full-Stack SaaS Platform (Next.js, MongoDB, NextAuth)

- Designed and shipped a full-stack web application end-to-end: requirements, data modeling, API design, and UI — giving users version-controlled resume repositories with commit history and rollback.
- Built a dual-render pipeline pairing a live HTML preview with a server-side compilation REST endpoint, demonstrating backend service design and API integration under real-time constraints.
- Modeled an append-only, audit-safe data schema (User → Repository → Version) — applying data-integrity and system-design principles relevant to enterprise application architecture.

Movie Recommendation System | Content-Based NLP Recommender Engine

- Built a content-based recommender over 5,000+ films using TF-IDF vectorization and cosine similarity, applying a full NLP preprocessing pipeline (tokenization, lemmatization, stop-word removal) to improve signal quality.
- Evaluated recommendation relevance using Precision@K (K = 5, 10, 20), enabling data-driven tuning of similarity thresholds and demonstrating a rigorous, metrics-based approach to testing and validation.

Tennis Ball Detection & Tracking System | Computer Vision Platform with REST API Integration

- Integrated a third-party REST API (Roboflow) into two production-quality tracking pipelines, handling external service integration, data transformation, and error handling.
- Built a TrackingAnalytics engine producing structured JSON output per session, and deployed the system as an interactive web UI with configurable runtime options and session history.

Next-Word Predictor | LSTM Language Model (TensorFlow/Keras)

- Trained an LSTM language model (Embedding → LSTM 150 units → Softmax Dense) on a 20,000-token corpus, building the full data pipeline: tokenization, sliding-window sequencing, padding, and one-hot encoding.
- Implemented checkpoint-based model persistence so training runs once and subsequent sessions load saved weights instantly — an optimization mindset applicable to maintaining stable, efficient production systems.

Web Prototype Generator | AI-Assisted Code Generation Tool (Flask backend)

- Led integration of a third-party API (Gemini) into a Flask backend service layer, building the request pipeline that turns natural-language input into functional HTML/CSS/JS output.
- Introduced in-source documentation and attribution standards adopted team-wide — reflecting a focus on maintainable, well-documented codebases across cross-functional collaboration.

WORK EXPERIENCE

Executive Assistant | Real Craft Tech Mar 2024 – Jun 2024

- Coordinated cross-functional priorities and communications for leadership, developing the stakeholder-facing communication and organizational skills that support collaborative, fast-moving engineering teams.
- Managed documentation and follow-through on assigned tasks with sincerity and consistency, reinforcing the detail-oriented habits now applied to code review, testing, and technical documentation.

Customer Operations and Marketing Executive | Real Craft Tech Mar 2023 – Jul 2023

- Handled customer-facing operations and marketing coordination, building the client communication and requirements-gathering instincts directly applicable to eliciting and documenting business requirements.
- Worked across operations and marketing functions, gaining early experience translating stakeholder needs into actionable follow-up — a precursor to translating business requirements into development tasks.

Website Design Intern | Rethink UX Oct 2022 – Dec 2022

- Took client-facing website features from requirements through delivery, building UI logic in HTML, CSS, and JavaScript and coordinating with the team on scope and integration.
- Implemented role-based access-control logic to govern feature visibility — experience directly applicable to securing REST endpoints and multi-tier application access.
- Wrote technical documentation for delivered features to support smooth team handoff and long-term maintainability.
- Participated in structured code reviews with senior engineers, incorporating feedback to meet production coding standards.

Programming Content Intern | StudyExperts Jun 2022 – Aug 2022

- Authored tutorials and knowledge-base articles on Python, Java, and data structures, translating technical concepts into clear, actionable guidance for end users.
- Wrote, tested, and debugged runnable code examples across multiple languages to guarantee published content accuracy — an early practice in rigorous testing and quality validation.
- Learned firsthand that strong documentation and product understanding are force multipliers on small teams, a principle now applied to project delivery and knowledge-base maintenance.

EDUCATION

Master of Science, Computer Science Aug 2024 – May 2026
University at Albany, State University of New York

Bachelor of Technology, Computer Science & Information Technology Aug 2020 – Apr 2024
Sri Indu College of Engineering & Technology (JNTUH)

CERTIFICATIONS & ACHIEVEMENTS

- Google AI Essentials — Google, July 2026
- Building with the Claude API — Anthropic, March 2026